

Mec645

Finite Element method in Fluid Mechanics

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Assignment No. 7: Convective problems and the GLS technique

Due in one week.

Concept question: What is the effect of the convection term on the local/global matrix?

Using SG and GLS techniques, solve the following non-dimensional convection-diffusion equation

$$P_e \frac{\partial T}{\partial x} = \frac{\partial^2 T}{\partial x^2}$$

where $P_e = \frac{uL}{\alpha}$ is the global Peclet number, with u is the convection speed, L is the domain length, and α is the thermal diffusivity. Given the following boundary conditions:

$$T(0) = 0, T(1) = 1$$

Get the solution for different Peclet numbers i.e. [1, 10, 100, 500].

Take: $\tau = \frac{1}{\sqrt{\left(\frac{2P_e}{L_e}\right)^2 + \left(\frac{4}{L_e^2}\right)^2}}$

Compare with the exact solution:

$$T(x) = \frac{e^{P_e x} - 1}{e^{P_e} - 1}$$

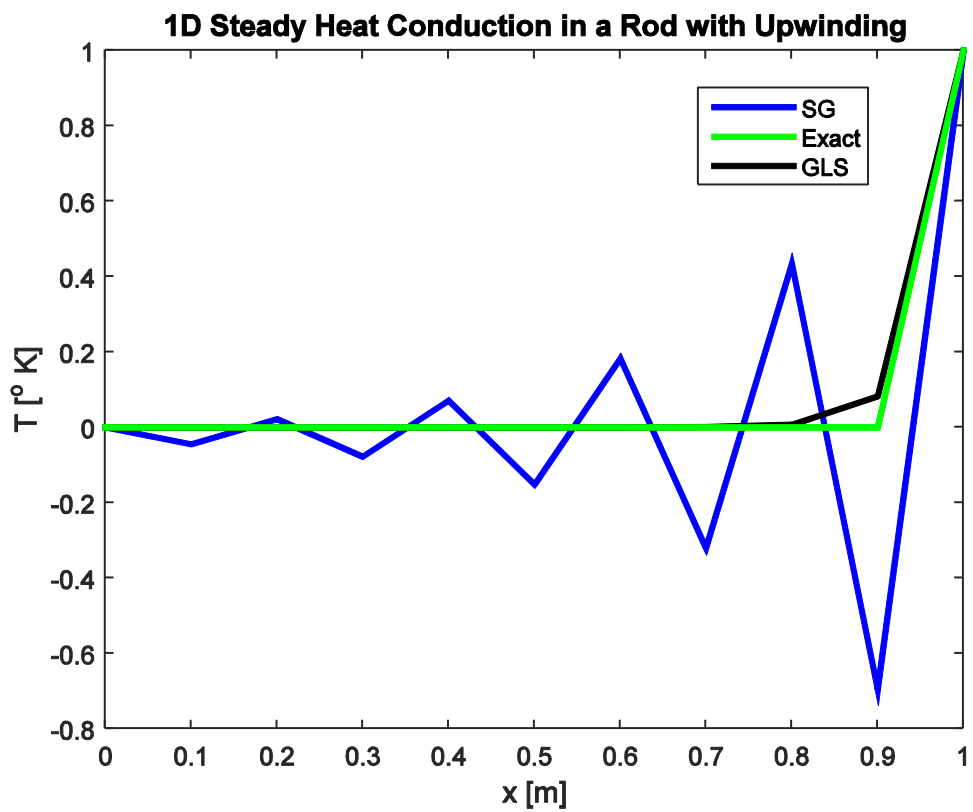
1. Plot the solution using the SG, GLS, and the exact solution for N=10 and 100 elements for the above range of the Pe . Then use clustering for N=10. And calculate the local Peclet number (based on the local values of its constituents e.g. L becomes L_e).
2. Calculate the scheme accuracy and CGI for the case of $Re = 500$.

For your reference, the general expression for the stabilization parameter is [1]:

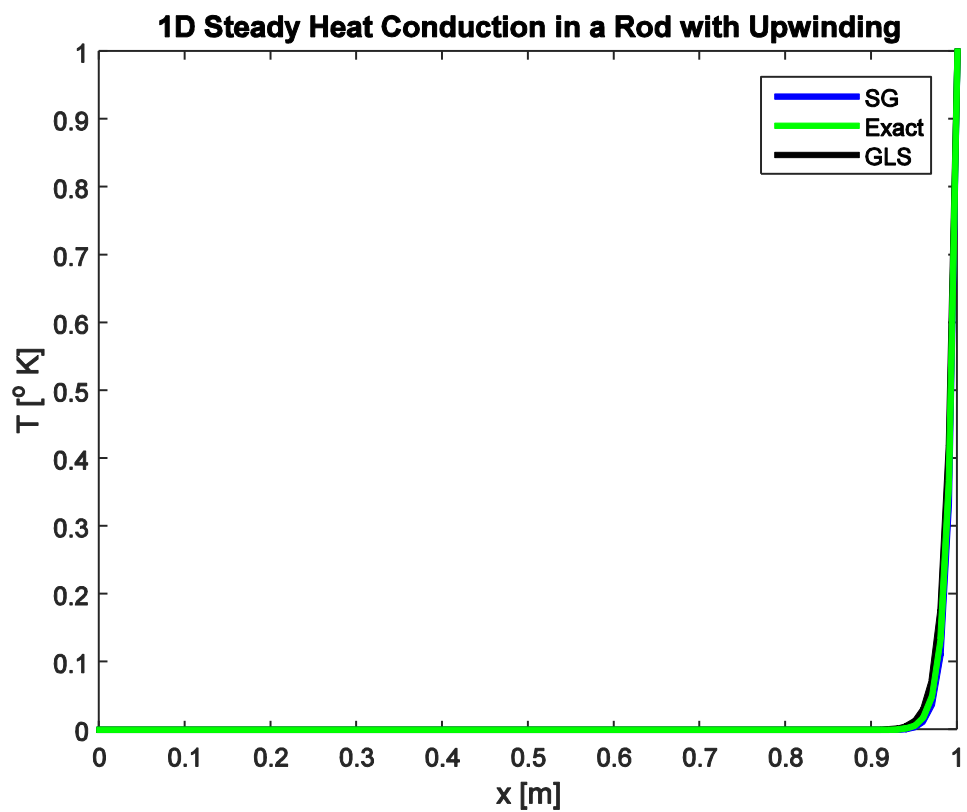
$$\tau = \frac{1}{\sqrt{\left(\frac{2}{\Delta t}\right)^2 + \left(\frac{2\|u^e\|}{L_e}\right)^2 + \left(\frac{4\alpha}{L_e^2}\right)^2}}$$

Sample Answers:

The case $Pe=100$



$N=10$ elements



$N=100$ elements

[1] T. JR. Hughes, L. P. Franca, and G. M. Hulbert. A new finite element formulation for computational fluid dynamics: Viii. the Galerkin/least-squares method for advective-diffusive equations. *Computer methods in applied mechanics and engineering*, 73(2):173-189, 1989.